PORTFOLIO · SELECTED WORK 2024–2026

# Davin Jonathan Tanus

## AI & Mobile Software Engineer — Information Systems, ITS

I build mobile and intelligent products end to end: a cross-platform asset-management app, an in-browser sign-language recognizer, an on-device-ready image classifier, a deployed RAG academic assistant, and rigorously evaluated ML systems. I enjoy challenge-based teamwork, turning user problems into shipped products with thoughtful experiences and measurable results.

Surabaya, Indonesia • **davinjtanus@gmail.com** • **+62 821-9465-5321** • **LinkedIn** • **GitHub** • **Portfolio**

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**Inside:** Profile & education · Featured mobile, AI & software projects with outcomes · Leadership & achievements · Technical skills · Selected certifications

## PROFILE

Information Systems undergraduate at Institut Teknologi Sepuluh Nopember (GPA 3.64/4.00) and AI Engineer Intern at PT Salam Pacific Indonesia Lines. My work spans **mobile development (React Native/Expo)**, full-stack engineering, **on-device-ready machine learning**, and applied AI products. I work in Agile teams, evaluate rigorously, and care about shipping products people actually use.

## EDUCATION

### Institut Teknologi Sepuluh Nopember (ITS)

**Bachelor's Degree in Information Systems · Aug 2023 – Present · GPA 3.64/4.00**

- Relevant areas: software engineering, data analytics, machine learning, IS/IT strategy, IT risk and service management.
- 2nd Place — Information Systems Programming Competition (HMSI ITS, 2024).

### AI Engineer Intern — PT Salam Pacific Indonesia Lines

**Performance Excellence Division · Applied AI & Full-Stack Engineering · 2026 – Present**

- Build internal document-intelligence and talent-development platforms (Next.js, FastAPI/Go, PostgreSQL/MySQL, Docker) with secure access flows, testing, and reporting.

## TECHNICAL SKILLS

**Mobile:** React Native / Expo (cross-platform app: registry, lifecycle tracking, role-based access, audit logs)

**AI & Machine Learning:** TensorFlow, PyTorch, scikit-learn, transfer learning, BiLSTM, time-series forecasting, on-device deployment (TensorFlow Lite, TensorFlow.js, SavedModel)

**Software & Data:** JavaScript, HTML, CSS, Python, SQL, Go · React, Next.js, FastAPI, Streamlit, REST APIs · PostgreSQL, MySQL, Supabase, MongoDB

**Cloud, DevOps & Ways of Working:** Power BI, Microsoft Fabric, Excel, PowerPoint · Docker, GCP + Cloud Run, GitHub Actions CI/CD, Git · Agile Scrum/Kanban, product-oriented delivery

## LEADERSHIP & ACHIEVEMENTS

### Scrum Master — Agile Snacklicious

**Scrum delivery · Six-person team · Four sprints**

- Led sprint execution, daily scrums, and retrospectives, divided tasks, and tracked income and expenses.

### Intern Staff — HMSI ITS (Information Systems Student Association)

**Sep 2024 – Dec 2024**

### 2nd Place — IS Programming Competition

**HMSI ITS · 2024**

## FEATURED PROJECTS — MOBILE & ON-DEVICE INTELLIGENCE

### AssetHub — Mobile Asset Management App GitHub | Case study

**Role: Mobile developer · Cross-platform (React Native / Expo)**

- **Problem:** enterprises need asset tracking beyond the desk — registry, lifecycle states, and accountability on the go.
- **Approach:** built searchable asset registry with filters, lifecycle tracking (Active, Maintenance, Borrowed), role-based access, and audit logs.
- **Outcome:** working cross-platform app delivered alongside a deployed web counterpart with shared access-control model.

React Native · Expo · Role-Based Access · Audit Logs

### Multi-Class Image Classification with On-Device Export Training notebook | Case study

**Role: ML engineer · Dicoding deep-learning project**

- **Problem:** accurate image recognition that can run beyond the lab — server, mobile, and browser.
- **Approach:** MobileNetV2 transfer learning on 13,394 Imagenette images (10 classes, 80/10/10 split), augmentation, staged fine-tuning, checkpointing; evaluated with learning curves, confusion matrix, per-class metrics.
- **Outcome:** 91.67% test accuracy; model exported to SavedModel, TensorFlow Lite (mobile), and TensorFlow.js (browser).

TensorFlow · MobileNetV2 · Transfer Learning · TFLite · TensorFlow.js

### RAG Akademik — AI Chatbot for Campus Regulations Live application | GitHub

**Role: Builder · Deployed product**

- **Problem:** students struggle to find answers across dozens of academic regulation documents.
- **Approach:** retrieval-augmented generation over 30+ indexed documents with semantic search and pre-computed embeddings for fast, grounded answers.
- **Outcome:** live assistant answering policy questions (SKS limits, thesis, GPA, scholarships, graduation).

RAG · LLM · Streamlit · Semantic Search

## FEATURED PROJECTS — ML RIGOR & ENGINEERING DEPTH

### BISINDO Fingerspelling — In-Browser Sign Language Recognition Live demo | GitHub

**Role: ML engineer & front-end · personal project · 26 letters, on-device**

- **Problem:** read the Indonesian Sign Language alphabet from a webcam without a server, and know how far the accuracy can be trusted.
- **Approach:** audited the public dataset (63% of validation images were near-copies of training images) and re-split by perceptual-hash groups; fine-tuned YOLO11n, chose on validation only, exported to ONNX and ran it client-side with ONNX Runtime Web plus an 8-frame vote.
- **Outcome:** 98.5% on a held-out test set (scored once) and 93.6% on an unseen signer; every error there falls on O or Y, and the failure analysis is published on the live demo page.

YOLO11 PyTorch ONNX Runtime Web Data Leakage Audit

### Indonesian App Review Sentiment Analysis Training notebook | Case study

**Role: ML engineer · Dicoding NLP project · 16,000+ self-scraped reviews**

- **Problem:** distill thousands of Indonesian app reviews into product signal (positive / neutral / negative).
- **Approach:** slang normalization, sentiment-aware preprocessing, deduplication, class balancing; benchmarked LinearSVC, logistic regression, and a PyTorch BiLSTM with per-class and confusion-matrix analysis.
- **Outcome:** 92.17% test accuracy with a repeatable inference module.

NLP Scikit-Learn PyTorch BiLSTM Text Classification

### Stock Price Forecasting — Classical ML, Deep Learning & Chronos-2

Classical notebook | Chronos-2 notebook

**Role: ML engineer · 10 years of UNTR.JK data**

- **Problem:** compare classical, deep, and foundation-model forecasting under realistic trading simulation.
- **Approach:** rolling walk-forward validation and residual diagnostics across ARIMA, Prophet, XGBoost, LSTM, and zero-shot Amazon Chronos-2 (120M).
- **Outcome:** custom LSTM at 1.04% MAPE; honest analysis of zero-shot limits (48.28% directional accuracy).

Time Series LSTM XGBoost Foundation Model

## SELECTED CERTIFICATIONS

**Belajar Fundamental Deep Learning — Dicoding Indonesia (Sep 2026)**
Computer vision, NLP, evaluation, deployment workflows

**Cisco DevNet Associate — Cisco Networking Academy × ITS (Jul 2026)**
APIs, automation, application security, testing, cloud deployment

**AWS AI Academy — Basic & Beginner — Dicoding × AWS (Aug 2026)**
AI foundations on cloud infrastructure

**LangChain Academy Foundations — LangChain, LangGraph, LangSmith (Aug 2026)**
LLM applications, agentic workflows, deployment operations

Full archive (24 certifications + learning activities with verification documents): **portofolio-liard-ten.vercel.app/credentials**

## MORE SELECTED WORK

Conduit CI/CD on Google Cloud (Docker, GitHub Actions, Cloud Run), AssetHub Web (React + Supabase, live), Academic Data Warehouse & ETL, BCA KPI dashboards, Banana-Quality clustering (K-Means + PCA), IT risk management, IS/IT strategic planning, and computer-vision simulation — all documented with reports and previews at **portofolio-liard-ten.vercel.app/project**.

> **Let's build something people love to use.**
> I learn fast, collaborate in Agile teams, and finish what I start — from model notebook to deployed product. Reach me at **davinjtanus@gmail.com** or **LinkedIn**.